

7th Heaven Cafe — Test Cases Specification Document

Complete Verification & Validation Test Matrix | System: Next.js 14, Cloud Firestore, Firebase Auth, Razorpay

Project: 7th Heaven Cafe (cafe 2)

Total Test Cases: 25

Pass Rate: 100% (25 Passed, 0 Failed)

Execution Date: September 2026

QA Lead: Aditya

TEST CASE ID	TEST CASE DESCRIPTION	TEST CATEGORY	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-01	Verify user registration with valid credentials	Registration	1. Navigate to /signup 2. Enter valid email and password 3. Click 'Sign Up' button	Email: user@example.com Password: Pass@1234	Account created in Firebase Auth; user is authenticated and redirected to home page /	As expected	Pass
TC-02	Verify registration fails with weak password (< 6 characters)	Registration	1. Navigate to /signup 2. Enter valid email and password with < 6 chars 3. Click 'Sign Up'	Email: test@domain.com Password: 123	Validation error displayed: password must be at least 6 characters; account creation rejected	As expected	Pass
TC-03	Verify user registration with an existing registered email	Registration	1. Navigate to /signup 2. Enter already registered email 3. Click 'Sign Up'	Email: alreadyregistered@test.com Password: Pass@1234	Error message 'Email already in use' displayed; registration denied	As expected	Pass
TC-04	Verify user login with valid credentials	Authentication	1. Navigate to /login 2. Enter valid registered email and password 3. Click 'Login'	Email: user@example.com Password: Pass@1234	Login succeeds; AuthContext updates with user session; redirected to home /	As expected	Pass
TC-05	Verify user login with invalid password	Authentication	1. Navigate to /login 2. Enter registered email with wrong password 3. Click 'Login'	Email: user@example.com Password: WrongPassword!	Authentication rejected; error banner 'Invalid email or password' displayed	As expected	Pass
TC-06	Verify Google OAuth single sign-on / signup	Authentication	1. Navigate to /login or /signup 2. Click 'Continue with Google' 3. Authorize Google account in popup	Authorized Google Account	User authenticates via Google popup; profile document initializes in Firestore; redirected to /	As expected	Pass
TC-07	Verify user logout functionality	Authentication	1. Log in with active user 2. Open Profile Drawer or Navbar menu 3. Click 'Logout'	Active session token	User session terminated, auth tokens revoked, navbar displays Login/Register buttons	As expected	Pass
TC-08	Verify menu browsing by category selection	Menu Browsing	1. Navigate to /menu 2. Click on category tabs ('Hot Coffee', 'Cold Coffee', 'Mug Cakes')	Selected Category: 'Cold Coffee'	Menu items instantaneously filter to display only items belonging to 'Cold Coffee'	As expected	Pass
TC-09	Verify live search by item name in menu	Menu Browsing	1. Navigate to /menu 2. Enter item keyword in search input field	Query: 'Vanilla Latte'	Grid filters dynamically in real-time to match 'Vanilla Latte' with correct pricing and tags	As expected	Pass
TC-10	Verify adding available menu item to cart	Cart Management	1. Navigate to /menu 2. Select an in-stock product card 3. Click 'Add to Cart'	Item: 'HazelNut Cold Coffee' (₹180)	Item added to CartContext; floating cart counter badge increments; notification shown	As expected	Pass
TC-11	Verify cart item quantity modification and price recalculation	Cart Management	1. Open Cart Drawer 2. Click '+' button to increment quantity 3. Click '-' button to decrement quantity	Item: 'Espresso', initial qty: 1 -> 2	Quantity changes dynamically; line total, 5% GST, ₹40 delivery fee, and grand total recalculate instantly	As expected	Pass

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TC-12	Verify removing individual item and clearing entire cart	Cart Management	1. Open Cart Drawer with 2 items 2. Click delete icon on item 1 3. Click 'Clear Cart'	Cart items: [Item A, Item B]	Item 1 is removed; clicking 'Clear Cart' resets items to empty state; 'Cart is Empty' banner shown	As expected	Pass
TC-13	Verify cart persistence across page reload	Cart Management	1. Add 2 items to cart 2. Hard refresh the page (Ctrl+F5 / Cmd+R) 3. Re-open Cart Drawer	Cart items in localStorage	Cart items, quantities, and calculated prices remain completely intact from localStorage	As expected	Pass
TC-14	Verify checkout prevention when cart is empty	Checkout	1. Ensure cart has 0 items 2. Direct URL access to /checkout or click checkout button	Empty cart: []	System displays alert 'Your cart is empty' and redirects user back to /#menu	As expected	Pass
TC-15	Verify checkout form auto-fills logged-in user profile	Checkout	1. Log in with user having saved address & phone 2. Navigate to /checkout	User profile with saved contact and address	Full Name, Phone Number, Email, and Delivery Address auto-populate the input fields	As expected	Pass
TC-16	Verify checkout form validation for mandatory delivery fields	Checkout	1. Open /checkout with items in cart 2. Clear or enter invalid name (< 2 chars) / phone (< 7 digits) / empty address 3. Click 'Proceed to Pay'	Name: 'A', Phone: '123', Address: ''	Form displays validation error alerts, smooth-scrolls to delivery section, blocks payment initialization	As expected	Pass
TC-17	Verify Razorpay payment modal invocation	Payment	1. Provide all valid contact and address details on /checkout 2. Click 'Proceed to Pay'	Valid customer form data, Cart total: ₹399	Razorpay modal pops up over the checkout page with order amount matching grand total	As expected	Pass
TC-18	Verify payment dismissal / failure handling	Payment	1. Open Razorpay checkout modal 2. Click close (X) or trigger payment failure	Dismissal action in payment popup	Modal closes gracefully; no order record created in Firestore; cart remains preserved with alert	As expected	Pass
TC-19	Verify successful order placement and Order Confirmation Card display	Order Placement	1. Complete successful payment transaction in Razorpay modal 2. Wait for success callback	Successful payment ID and verified signature	Order saved to Firestore with sequential Order ID; Cart cleared; Order Confirmation Card shown with ETA (25 mins)	As expected	Pass
TC-20	Verify Invoice Modal generation with complete pricing breakdown	Invoicing	1. On Order Confirmation screen, click 'View Invoice' button	Placed Order ID: #7H-1042	Invoice modal opens displaying date, customer details, itemized list, 5% GST, delivery fee, and payment method	As expected	Pass
TC-21	Verify real-time order tracking status updates	Order Tracking	1. Open /track-order/[orderId] 2. Update status in backend (Placed -> Preparing -> Out for Delivery -> Delivered)	Live order document in Firestore	Status timeline and visual tracker update instantly in real-time via Firestore snapshot without page reload	As expected	Pass
TC-22	Verify Kitchen Display queue ordered First-Come-First-Served (FCFS) in Admin	Admin Panel	1. Log in to /admin 2. Inspect incoming orders queue	Multiple active orders placed at different timestamps	Orders are strictly sorted in chronological order (oldest to newest) to maintain fair kitchen fulfillment	As expected	Pass
TC-23	Verify admin stock toggle reflects immediately on customer menu	Stock Control	1. In /admin, toggle 'inStock' switch for 'Belgian Mug Cake' to false 2. Switch to customer /menu view	Item: 'Belgian Mug Cake'	Item immediately renders with 'Out of Stock' badge; 'Add to Cart' button is disabled	As expected	Pass

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TC-24	Verify customer profile address update and synchronization	User Profile	1. Open Profile Drawer 2. Enter new delivery address & updated phone number 3. Click 'Save Profile'	Address: 'Flat 402, Sunshine Heights, Pune', Phone: '9876543210'	Details saved to user's Firestore document; updated address appears in checkout address choices	As expected	Pass
TC-25	Verify responsive UI layouts across mobile viewport (375px)	Responsive Design	1. Open browser dev tools in mobile emulation mode (375x667px) 2. Navigate through Navbar, Menu, Cart Drawer, and Checkout	Mobile viewport width: 375px	Navigation collapses to hamburger menu; Cart Drawer slides full width; checkout columns stack without overflow	As expected	Pass